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# -*- coding: utf-8 -*-
"""DEEP LEARNING EX 1

Automatically generated by Colab.

Original file is located at
https://colab.research.google.com/drive/1\_4W93X\_1Xdtate-7GHLz4fx4\_zWlozDO
"""

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler
from tensorflow.keras.models import Sequential # Space was missing between "models" and "import"
from tensorflow.keras.layers import Dense
from google.colab import auth
import gspread
from google.auth import default

auth.authenticate_user()
creds, _ = default()
gc = gspread.authorize(creds)
worksheet = gc.open('DEEP LEARNING ').sheet1
data = worksheet.get_all_values()

dataset1 = pd.DataFrame(data[1:], columns=data[0])
dataset1 = dataset1.astype({'X': 'float'})
dataset1 = dataset1.astype({'Y': 'float'})
dataset1.head()

X = dataset1[['X']].values
y = dataset1[['Y']].values
#X
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.33, random_state = 33)
Scaler = MinMaxScaler()
Scaler.fit(X_train)

X_train1 = Scaler.transform(X_train)
ai_brain = Sequential ([
Dense (8, activation = 'relu'),
Dense (10, activation = 'relu'),
Dense (1)
])
ai_brain.compile(optimizer = 'rmsprop', loss = 'mse')
ai_brain.fit(X_train1, y_train, epochs =3000)

# KRISHNARAJ D
loss_df= pd.DataFrame(ai_brain.history.history)
loss_df.plot()

X_test1=Scaler.transform(X_test)
ai_brain.evaluate(X_test1, y_test)

X_n1=[[7]]
X_n1_1=Scaler.transform(X_n1)
ai_brain.predict(X_n1_1)

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